

VINJEORA II, Norway

WMO: 012320

Lat: 63.200N

Lon: 9.000E

Elev: 67

StdP: 100.52

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-9.8	-18.5	0.7	-10.1	-16.8	0.9	-8.1	10.1	2.5	9.2	1.4	4.3	170	0.596

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	24.6	15.9	22.6	14.9	21.0	14.2	17.4	22.4	16.2	20.7	15.2	19.2	3.0	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	11.0	19.4	14.3	10.3	18.1	13.4	9.6	16.9	49.2	22.3	45.7	20.9	42.9	19.1	22.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.0	6.9	6.1	-12.6	28.6	1.7	2.6	-13.9	30.4	-14.9	31.9	-15.9	33.4	-17.1	35.2		
(5)			-13.6	19.1	1.9	1.5	-14.9	20.1	-16.0	21.0	-17.0	21.8	-18.4	22.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-0.9	0.4	2.0	6.1	10.3	12.4	15.2	14.5	11.5	6.6	3.2	1.0		
	DBStd	6.75	4.73	4.45	3.78	3.78	4.13	2.79	3.29	2.77	3.28	4.01	4.35	4.50		
	HDD10.0	1668	339	268	249	127	47	9	1	1	21	120	207	280		
	HDD18.3	4196	597	501	507	367	251	178	109	123	204	365	454	538		
	CDD10.0	531	0	0	1	9	56	81	161	141	67	13	3	0		
	CDD18.3	18	0	0	0	0	1	1	11	4	1	0	0	0		
	CDH23.3	129	0	0	0	0	12	7	89	21	0	0	0	0		
	CDH26.7	22	0	0	0	0	1	0	19	1	0	0	0	0		
(14) Wind	WSAvg	2.7	3.8	3.5	2.9	2.5	2.4	2.0	1.8	1.8	2.1	2.8	3.1	3.4		
(15) Precipitation	PrecAvg	1495	146	138	142	79	78	108	101	131	146	137	133	155		
	PrecMax	1891	407	312	355	220	164	208	169	294	412	271	355	343		
	PrecMin	1004	4	3	35	17	45	42	30	45	51	27	4	36		
	PrecStd	217	93	72	86	52	27	43	38	60	87	51	98	71		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.1	12.7	13.7	18.9	24.7	24.1	29.3	25.7	21.7	18.7	16.2	12.3		
		MCWB	5.0	6.3	6.9	10.0	14.3	14.8	19.2	15.0	14.0	11.1	9.6	6.4		
	2%	DB	8.6	9.6	10.6	16.7	22.2	21.4	25.6	23.1	19.7	16.4	13.5	10.2		
		MCWB	3.8	4.2	5.0	8.9	13.2	13.5	17.3	14.7	13.1	10.3	8.1	4.8		
	5%	DB	7.2	8.1	9.0	14.4	20.2	19.7	23.4	21.3	18.2	14.2	11.2	8.5		
		MCWB	2.8	3.2	4.1	7.8	12.0	13.0	16.2	14.4	12.2	8.6	6.2	4.0		
	10%	DB	5.1	6.6	7.5	12.2	17.6	18.1	21.3	19.7	17.1	12.6	9.2	6.9		
		MCWB	1.6	2.2	3.4	6.4	10.9	12.1	15.6	14.0	11.9	7.6	5.3	3.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.4	6.6	7.3	10.9	15.6	16.2	20.1	17.5	15.9	12.5	9.7	7.1		
		MCDB	9.4	12.0	11.6	17.6	23.7	21.5	26.9	21.9	19.4	16.4	15.5	11.0		
	2%	WB	4.1	4.6	6.0	9.2	13.9	14.6	18.2	16.3	14.2	10.7	8.3	5.7		
		MCDB	8.0	8.7	9.4	15.4	20.1	19.0	23.6	20.4	17.9	15.2	12.6	9.1		
	5%	WB	3.0	3.6	4.9	8.2	12.5	13.6	17.2	15.5	13.2	9.4	7.1	4.5		
		MCDB	6.5	7.2	8.3	13.6	18.5	18.1	22.3	19.2	16.9	13.5	11.0	7.8		
	10%	WB	2.0	2.6	3.8	7.0	11.4	12.7	16.1	14.7	12.4	8.2	5.5	3.4		
		MCDB	4.5	5.8	6.6	11.5	17.1	16.9	20.3	18.2	15.8	11.9	8.5	6.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.9	6.3	8.2	9.0	8.3	8.2	7.8	7.1	5.6	4.7	4.8		
		MCDBR	6.6	6.6	8.1	10.8	12.8	11.8	12.4	10.3	8.6	7.5	7.1	7.4		
	5% WB	MCWBR	4.0	3.8	4.6	5.3	5.6	5.6	5.5	4.4	4.2	3.9	4.1	4.5		
		MCWBR	6.5	6.7	7.9	10.2	11.8	10.1	11.5	8.9	8.5	7.1	6.5	7.3		
(40) Clear-Sky Solar Irradiance	taub	taub	0.244	0.302	0.333	0.360	0.360	0.362	0.371	0.358	0.337	0.312	N/A	0.365		
		taud	2.031	2.185	2.225	2.282	2.362	2.424	2.433	2.468	2.504	2.396	N/A	2.541		
	Edn at Noon	Edn at Noon	363	620	753	817	853	859	839	819	766	623	N/A	191		
		Edh at Noon	31	63	89	105	105	101	98	87	69	51	N/A	18		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.16	0.69	1.98	3.71	4.80	5.01	4.64	3.57	2.15	0.95	0.25	0.08		
	RadStd	RadStd	0.03	0.11	0.26	0.26	0.26	0.43	0.40	0.35	0.24	0.13	0.04	0.01		

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-143	-189	+67	N/A	(47)

Nomenclature: See separate page